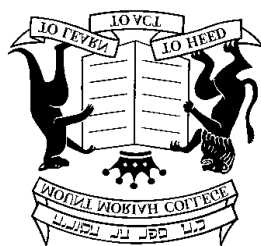


Student Number _____



MORIAH COLLEGE

Year 12 – Task 2 - Pre-Trial

MATHEMATICS 2009

Time Allowed: 3 hours

Examiners: C. White, L. Bornstein

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–10
- All questions are of equal value

Question 1. (12 Marks) Use a SEPARATE Answer Sheet.

Marks

- (a) Evaluate correct to 3 significant figures: **2**

$$\frac{e^3 + 2.58}{\sqrt{3.5 - 2.3}}$$

- (b) Solve $|x - 5| \leq 7$ **2**

- (c) Solve $x^2 - 3x - 10 = 0$ **2**

- (d) Differentiate $x^6 - 2x^{-1} + 3$ with respect to x **2**

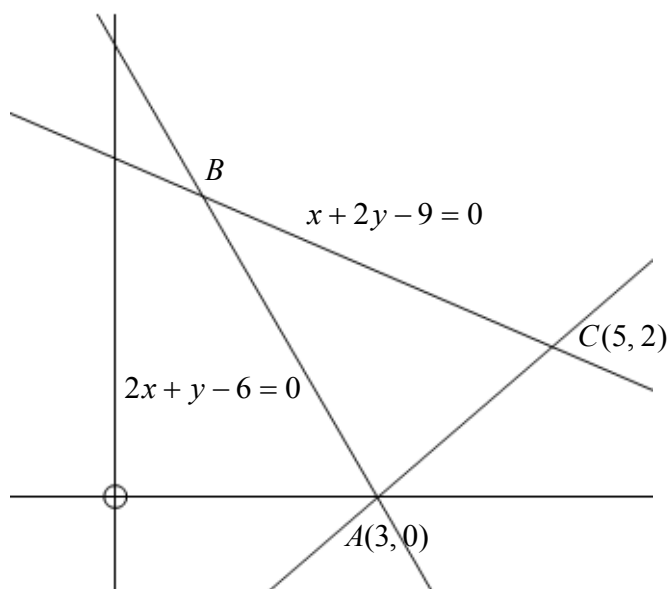
- (e) Find integers a and b such that $(4 + \sqrt{3})^2 = a + b\sqrt{3}$ **2**

- (f) Solve the following pair of simultaneous equations: **2**

$$\begin{aligned} 3x + 2y &= 14 \\ 2x - 4y &= -12 \end{aligned}$$

- (a) Find the equation of the normal to the curve $y = x^2 + 5x$ at the point $(1, 6)$. 3

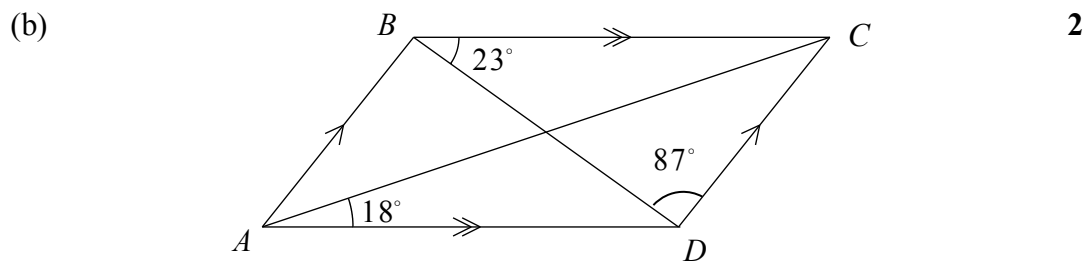
(b)



The points $A(3, 0)$, B and $C(5, 2)$ form a triangle, as shown in the diagram. The line BC has equation $x + 2y - 9 = 0$ and the line AB has equation $2x + y - 6 = 0$.

- (i) Find the gradient of the line AC . 1
- (ii) Show that the equation of the line AC is $x - y - 3 = 0$. 1
- (iii) Show that the coordinates of the point B are $(1, 4)$. 2
- (iv) Show that the perpendicular distance between the point B and the line AC is $3\sqrt{2}$. 2
- (v) Find the area of the triangle ABC . 3

(a) Evaluate $\int_1^2 \left(\frac{3x^4 - 2x}{x^3} \right) dx$ 2

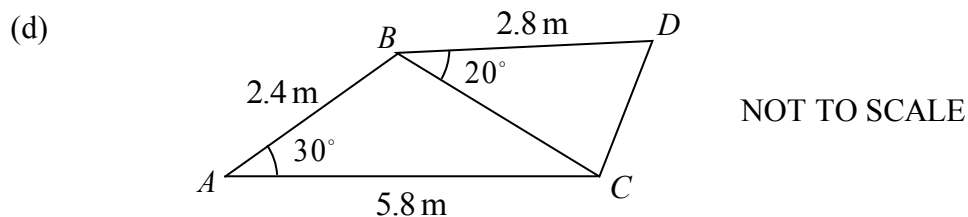


In the diagram $ABCD$ is a parallelogram. Given that $\angle CBD = 23^\circ$, $\angle CAD = 18^\circ$ and $\angle BDC = 87^\circ$, find the size of $\angle ACD$. Give reasons for your answer.

(c) Differentiate with respect to x :

(i) $(x^2 - 3)^7$ 2

(ii) $\frac{2x}{e^x}$ 2



In the diagram above $AB = 2.4$ m, $AC = 5.8$ m, $BD = 2.8$ m, $\angle BAC = 30^\circ$ and $\angle DBC = 20^\circ$.

(i) Show that the length of $BC = 3.9$ m, to 1 decimal place. 2

(ii) Hence or otherwise find the area of triangle BCD , to the nearest metre. 2

(a) The first 3 terms of arithmetic series are $-5+1+7+\dots$:

(i) Find the 55th term of the series. 2

(ii) Hence, or otherwise find the sum of the first 55 terms of the series. 2

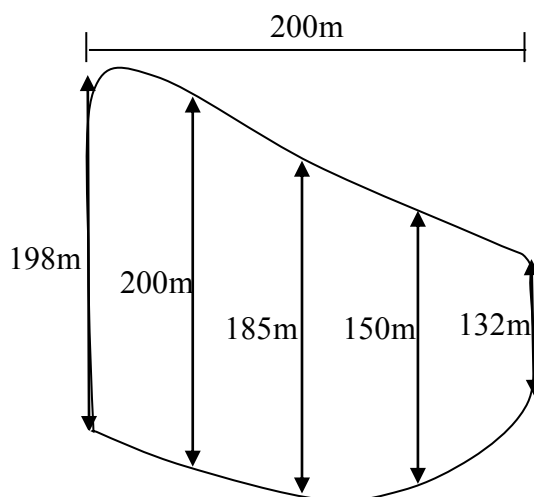
(b) Given that $\sin \theta = \frac{2}{3}$ and $90^\circ \leq \theta \leq 180^\circ$, find the exact value of the 3
following expressions, making sure that you fully simplify your answers:

i) $\cos \theta$.

ii) $\sin \theta - \tan \theta$.

(c) State the domain and range of the function $y = x^2 - 9$. 2

(d) The diagram below shows an aerial view of the Duck Pond in 3
Centennial park. The length of the Duck Pond is 200 m and the width
of the lake is shown at 50 metre intervals.



Use Simpson's rule with 5 function values to find an approximation for the surface area of the lake.

- (a) Find the values of k for which the quadratic equation
 $0 = x^2 - kx + k + 8$:

(i) has two equal roots. **1**

(ii) has two real roots. **2**

- (b) Find all the values of θ such that $\sin 2\theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta \leq 360^\circ$ **3**

- (c) (i) Sketch the curve $y = x^2 - 3x + 2$, clearly indicating the x and y intercepts. **2**

- (ii) On the same set of axes from part (i), sketch the line $y = 2 - x$ and shade the region represented by: **2**

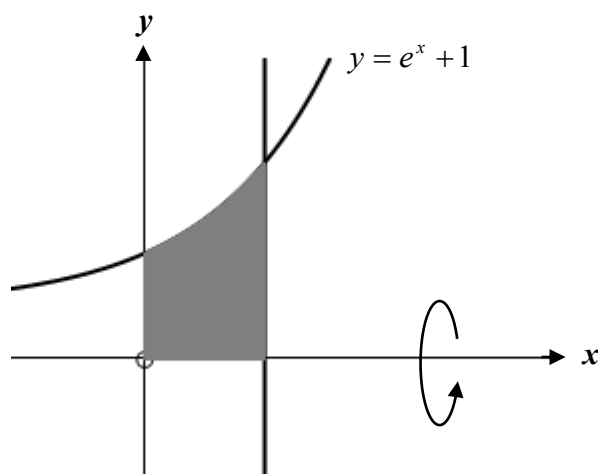
$$\int_0^2 (2 - x - (x^2 - 3x + 2)) dx$$

- (iii) Find the exact value of the area of the shaded region in part (ii) **2**

- (a) A function $f(x)$ is defined by $f(x) = x^3 - 6x^2 + 9x - 4$ for the domain $-1 \leq x \leq 5$.
- (i) Find the coordinates of the turning points of $f(x)$ and determine their nature. 3
- (ii) Find the coordinates of the point of inflexion. 2
- (iii) Hence, sketch the graph of $y = f(x)$, showing all turning points, the point of inflexion and the y -intercept. 3

- (b) Evaluate $\sum_{r=3}^6 \frac{1}{r^2}$, to 2 decimal places. 1

- (c) 3



In the diagram the shaded region is bounded by the curve $y = e^x + 1$, the x -axis, the y -axis and the line $x = 1$.

Find the exact volume of the solid formed when the region is rotated about the x -axis, making sure that you fully simplify your answer.

(a) Let α and β be the solutions to the equation $0 = x^2 - 5x + 7$.

(i) Find $\alpha\beta$ and $\alpha + \beta$. **1**

(ii) Hence, find $\alpha^3\beta + \alpha\beta^3$. **2**

(b) A parabola is defined by the equation $-8x = y^2 - 4y - 4$.

(i) Show that the equation can be written as $(y - 2)^2 = -8(x - 1)$. **1**

(ii) Sketch the parabola, clearly indicating the coordinates of the vertex, the coordinates of the focus and the equation of the directrix. **4**

(c) Consider the geometric series

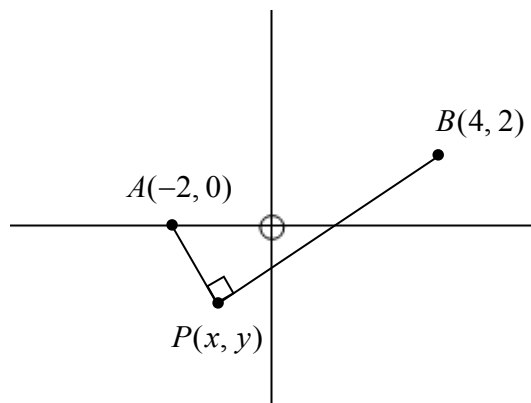
$$1 + (\sqrt{3} - 2) + (\sqrt{3} - 2)^2, \dots$$

(i) Find the value of the 7th term of the series to 2 significant figures. **1**

(ii) Explain why the series has a limiting sum. **1**

(iii) Find the exact value of the limiting sum of the series, making sure that you fully simplify your answer. **2**

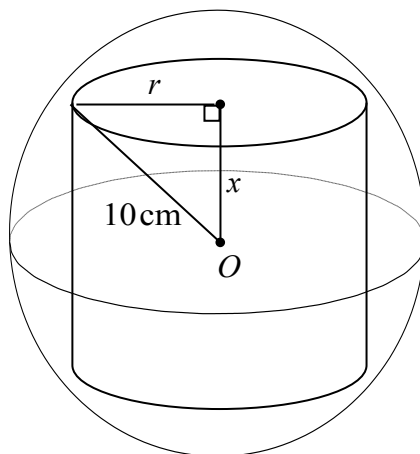
(a)



In the diagram above the points A , B , and P have coordinates $(-2, 0)$, $(4, 2)$ and (x, y) respectively.

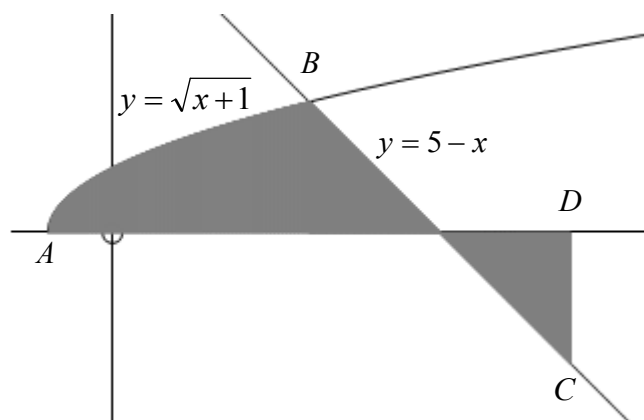
- (i) Given that $\angle APB = 90^\circ$, show that the locus of the point P is the curve with equation $x^2 - 2x + y^2 - 2y - 8 = 0$. 2
 - (ii) Show that this is a circle and find its centre and radius. 2
- (b) Solve the equation $3^{2x} - 7(3^x) - 18 = 0$. 2
- (c) Find the coordinates of the point P on the curve $y = \frac{1}{(x-5)^2}$ at which the tangent to the curve is perpendicular to the line $y = 4x - 3$. 3
- (d) Differentiate $3xe^x$ and hence find $\int_0^2 e^x + xe^x dx$. 3

(a)



A cylinder is to be cut out of a sphere, as shown in the diagram.
The sphere has radius 10 cm and the cylinder has radius r cm and height $2x$ cm.

- (i) Show that the volume, V , of the cylinder is $V = 2\pi x(100 - x^2)$ **2**
 - (ii) Find the value of x for which the volume of the cylinder is a maximum. You must give reasons why your value of x gives the maximum volume. **3**
- (b) Find $f(x)$, if $f'(x) = e^{3x-2}$ and $f(1) = \frac{4e}{3}$. **2**
- (c) The shaded region $ABCD$ is bounded by the line $x = 7$, the curve $y = \sqrt{x+1}$, the line $y = 5 - x$ and the x -axis, as in the diagram.

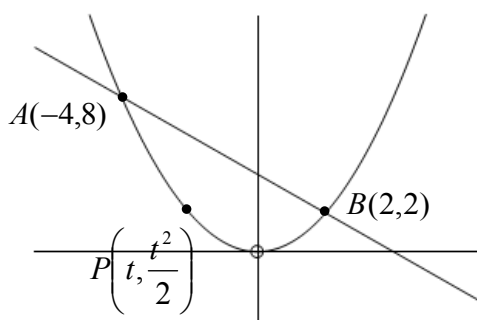


- (i) Show that B has coordinates $(3, 2)$. **2**
- (ii) Find the shaded area. **3**

(a) (i) Prove that $(1 + \tan^2 \theta) \sin^2 \theta = \tan^2 \theta$. **2**

(ii) Hence, or otherwise solve the equation **3**
 $2(1 + \tan^2 \theta) \sin^2 \theta + 5 \tan \theta = 7$ for $-180^\circ \leq \theta \leq 180^\circ$, giving
 your answer to the nearest minute.

(b)



In the diagram $A(-4, 8)$ and $B(2, 2)$ are points of intersection of the parabola $y = \frac{x^2}{2}$ with the line $y = 4 - x$. The point $P\left(t, \frac{t^2}{2}\right)$ is a variable point on the parabola, below the line.

(i) Find the area between the line $y = 4 - x$ and the curve $y = \frac{x^2}{2}$. **3**

(ii) Show that the shortest distance between the line and the point P **1**
 can be given by the expression:

$$d = \frac{4 - t - \frac{t^2}{2}}{\sqrt{2}} \text{ for } -4 \leq t \leq 2.$$

(iii) Hence, show that the maximum area of the triangle APB is $\frac{3}{4}$ of the **3**
 area between the line and the curve.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

#1a) 20.7 (to 3sf).



$$\begin{aligned} b) \quad x-5 &\leq 7 \quad \text{AND} \quad x-5 \geq -7 \quad \checkmark \\ x &\leq 12 \quad \quad \quad x \geq -2. \\ -2 &\leq x \leq 12. \quad \checkmark \end{aligned}$$

$$\begin{aligned} c) \quad (x-5)(x+2) &= 0 \quad \checkmark \\ x &= 5 \quad x = -2. \quad \checkmark \end{aligned}$$

$$\begin{aligned} d) \quad y &= x^6 - 2x^{-1} + 3 \\ \frac{dy}{dx} &= 6x^5 + 2x^{-2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} e) \quad 16 + 9\sqrt{3} + 3 &= a + b\sqrt{3} \\ \therefore a &= 19 \quad \checkmark \\ b &= 8. \quad \checkmark \end{aligned}$$

$$\begin{aligned} f) \quad 6x + 4y &= 28 \\ -8x + 12y &= +36 \\ \hline 16y &= 64 \\ y &= 4 \quad \checkmark \\ 3x + 8 &= 14 \\ 3x &= 6 \\ x &= 2. \quad \checkmark \\ \hline &0. \end{aligned}$$

2a). $\frac{dy}{dx} = m(\tan) = 2x+5$
at $x=1$

$m(\tan) = 7$ ✓

$m(\text{normal}) = -\frac{1}{7}$ ✓ (1, 6)

∴ eq: $y - 6 = -\frac{1}{7}(x - 1)$ 3.
 $7y - 42 = -x + 1$
 $7y = -x + 43$ ✓

b) $m(AC) = \frac{2-0}{5-3} = \frac{2}{2} = 1$ ✓ 1

ii) AC: $y - 2 = 1(x - 5)$

$y - 2 = x - 5$

$y = x - 3$ ✓ 1

$x - y - 3 = 0$

iii) $x = 9 - 2y$ — ①

$2(9 - 2y) + y - 6 = 0$

$18 - 4y + y - 6 = 0$ 2

$-3y = -12$

$y = 4$ ✓ ②

$x = 9 - 8$

$x = 1$ ✓ B(1, 4)

iv) B(1, 4) $\begin{matrix} a & b & c \\ |x & + & y & - & 3 & = & 0 \end{matrix}$

$\perp d = \frac{|(1)(1) + (-1)(4) + (-3)|}{\sqrt{1^2 + (-1)^2}} = \frac{|1 - 4 - 3|}{\sqrt{2}}$ ✓

$= \frac{6 \times \sqrt{2}}{\sqrt{2} \sqrt{2}}$

$= \frac{6\sqrt{2}}{2}$ ✓ 2

$= 3\sqrt{2}$ ✓

$d(AC) = \sqrt{(5-3)^2 + (2-4)^2}$

$= \sqrt{4 + 4}$

$= \sqrt{8}$ ✓

∴ v) area of $\Delta ABC = \frac{1}{2}(\sqrt{8}(3\sqrt{2})) = 6u^2$ ✓ 2

3a). $\int_1^2 \frac{3x^4}{x^3} - \frac{2x}{x^3} dx = \int_1^2 3x - 2x^{-2}$
 $= \left[\frac{3x^2}{2} - \frac{2x^{-1}}{-1} \right]_1^2$ ✓
 $= \left[\frac{3x^2}{2} + \frac{2}{x} \right]_1^2$ 2
 $= (6+1) - \left(\frac{3}{2} + 2 \right)$
 $= (7) - (3\frac{1}{2})$ ✓
 $= \underline{3\frac{1}{2}}$

b). $\angle BDA = 23^\circ$ alt $\angle =$
 $\therefore \angle BCD = 70^\circ$ const $\angle$ suppl. ✓ 2.
 $\angle ACD = 52^\circ$ ✓

k) i) $y = (x^2 - 3)^7$
 $y' = 7(x^2 - 3)^6 \times 2x \checkmark = 14x(x^2 - 3)^6 \checkmark$ 2.
 ii) $y = \frac{2x}{e^x}$

$u = 2x$ $v = e^x$ 2.
 $u' = 2$ $v' = e^x$

$y' = \frac{(e^x)(2) - (2x)(e^x)}{(e^x)^2} \checkmark = \frac{2e^x(1-x)}{e^{2x}} \checkmark = \frac{2(1-x)}{e^x}$

d) i) $BC^2 = 2.4^2 + 5.8^2 - 2(2.4)(5.8)\cos 30^\circ \checkmark$
 $BC^2 = 15.289$
 $BC = \underline{3.9}$ ✓ 2.

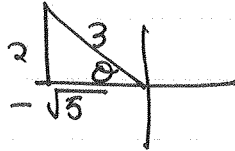
ii) $\text{area} = \frac{1}{2}(2.8)(3.9)\sin 20^\circ \checkmark$
 $= 1.867$ 2.
 $= 2 \text{ m}^2 \text{ (to nearest m)}$
 ✓

#4a) $-5 + 1 + 7 \dots$ $a = -5$, $d = 6$ ✓

i) $T_{55} = a + 54d$
 $= -5 + 54(6)$
 $= 319$ ✓

ii) $S_{55} = \frac{55}{2} [2(-5) + (54)(6)]$ ✓
 $= 8635$ ✓

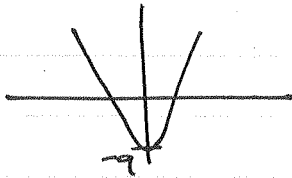
b) $\sin \theta = \frac{2}{3} \frac{p}{h}$



i) $\cos \theta = -\frac{\sqrt{5}}{3}$ ✓

ii) $\sin \theta - \tan \theta = \frac{2}{3} - \frac{2}{-\sqrt{5}} \frac{\sqrt{5}}{\sqrt{5}} = \frac{2}{3} + \frac{2\sqrt{5}}{5}$ ✓ (simplified correctly)

c)



D: $x \in \mathbb{R}$ ✓

R: $y \geq -9$ ✓

d).

x	y	δ	δxy
0	198	1	198
50	200	4	800
100	185	2	370
150	150	4	600
200	132	1	132

$\Sigma 2100$ ✓

$\therefore \text{Area} = \frac{50 \times 2100}{3}$
 $= 35000 \text{ m}^2$ ✓

#5 a). for equal roots $\Delta = 0$

$$(-k)^2 - 4(1)(k+8) = 0$$

$$k^2 - 4k - 32 = 0$$

$$(k-8)(k+4) = 0$$

$$k=8 \quad k=-4$$

for real roots $\Delta \geq 0$



$$k \leq -4$$

$$k \geq 8$$

(forget equals sign then 1 mark only)

b) $\sin 2\theta = -\frac{\sqrt{3}}{2}$

rel $\angle = 60^\circ$

$\sin$ is - in

3rd	4th
-----	-----

$$2\theta = 240^\circ, 600^\circ$$

$$\theta = 120^\circ, 300^\circ$$

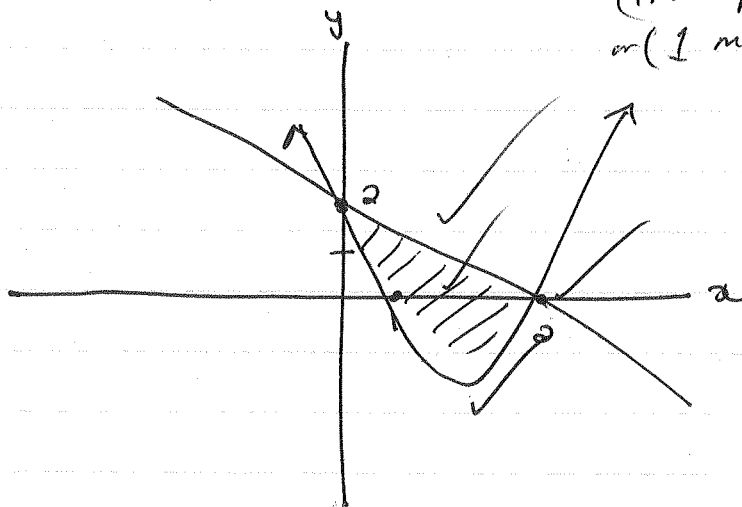
$$2\theta = 300^\circ, 840^\circ$$

$$\theta = 150^\circ, 330^\circ$$

(correct domain)

(1 mark for partial soln,
or (1 mark for rel. $\angle$)

c) $y = x^2 - 3x + 2$
 $y = (x-2)(x-1)$



$$A = \int_0^2 (2-x-x^2+3x-2) dx$$

$$= \int_0^2 2x - x^2 dx$$

$$= \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_0^2$$

$$= (4 - \frac{8}{3}) - (0)$$

$$= \frac{4}{3} u^2$$

#6. $y = x^3 - 6x^2 + 9x - 4$ $-1 \leq x \leq 5$

S.P. $y' = 0$: $3x^2 - 12x + 9 = 0$ $\frac{6}{3 \cdot 2}$
 $(x-1)(x-3) = 0$
 $x = 1$ | $x = 3$ ✓
 $y = 0$ | $y = -4$ ✓

note: $y'' = 6x - 12$
 at $x = 1$
 $y'' = -6$
 $\curvearrowright$ max.
 $(1, 0)$ max

at $x = 3$
 $y'' = 6$
 $\curvearrowleft$ min
 $(3, -4)$ min ✓

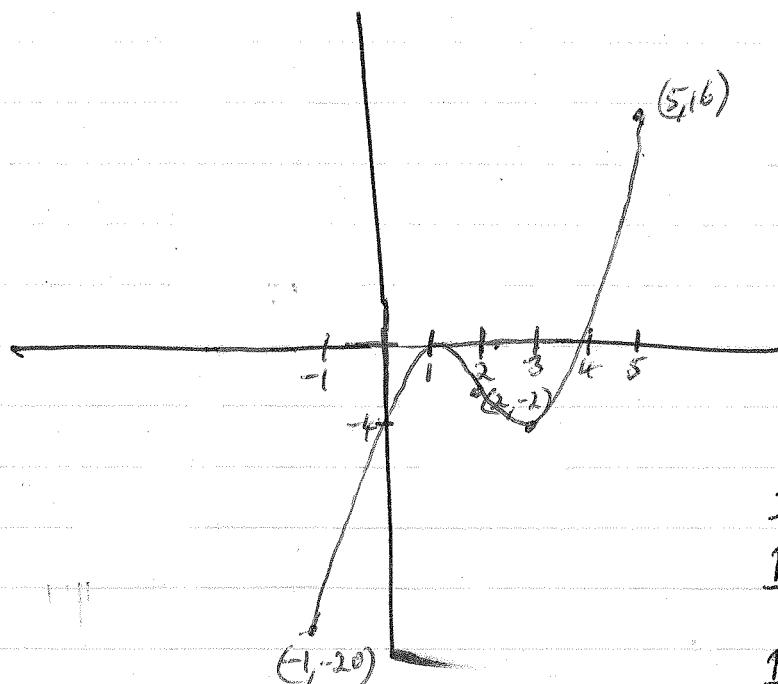
P.O.I. $y'' = 0$
 $x = 2$
 $y = -2$ ✓

check change in concavity (or similar)

x	0	2	3
y''	-12	0	6

$x = -1$
 $y = -20$

$x = 5$
 $y = 16$



1 - end points

1 - max, min & pt

1 - shape & direction of curve.

#6b) $\frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \frac{1}{36} = 0.24$ (to 2 dp).

c)

$$V = \pi \int_0^1 (e^x + 1)^2 dx$$

$$= \pi \int_0^1 e^{2x} + 2e^x + 1 dx$$

$$= \pi \left[\frac{e^{2x}}{2} + 2e^x + x \right]_0^1$$

$$= \pi \left[\left(\frac{e^2}{2} + 2e + 1 \right) - \left(\frac{1}{2} + 2 \right) \right]$$

$$= \pi \left(\frac{e^2}{2} + 2e + 1 - 2\frac{1}{2} \right)$$

$$= \pi \left(\frac{e^2}{2} + 2e - 1\frac{1}{2} \right) u^3.$$

#7a). $x^2 - 5x + 7 = 0$

$$\alpha + \beta = 5$$

$$\alpha\beta = 7$$

ii) $\alpha^3\beta + \alpha\beta^3$

$$= \alpha\beta(\alpha^2 + \beta^2)$$

$$= \alpha\beta((\alpha + \beta)^2 - 2\alpha\beta)$$

$$= 7(5^2 - 2 \times 7)$$

$$= 7(25 - 14)$$

$$= 77$$

7b). $-8x + 4 = y^2 - 4y$

$$-8x + 4 + (2)^2 = y^2 - 4y + (2)^2$$

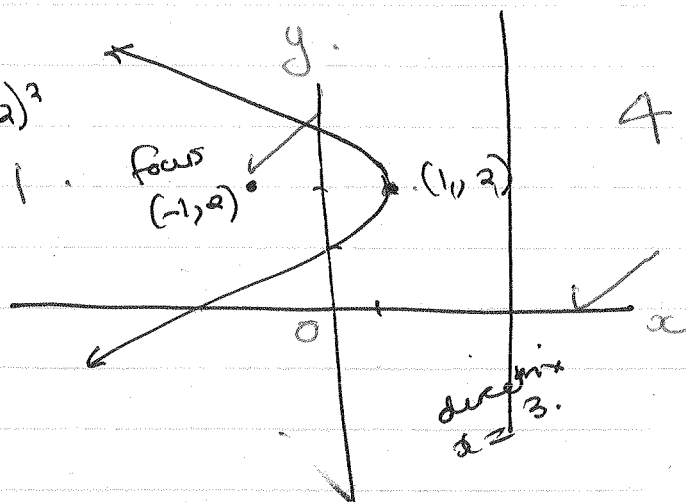
$$(y-2)^2 = -8x + 8$$

$$(y-2)^2 = -8(x-1)$$

$$(y-2)^2 = -4(2)(x-1)$$

$$\therefore \text{vertex } (1, 2)$$

$$\text{focal length} = 2.$$



$$7c) i). 1 + (\sqrt{3}-2) + (\sqrt{3}-2)^2 \dots$$

$$T_7 = ar^6$$

$$= 1(\sqrt{3}-2)^6$$

$$= 0.00637 \checkmark$$

$$ii) r = \sqrt{3}-2 \approx -0.267$$

$$-1 < r < 1 \checkmark$$

$$iii) S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1}{1-(\sqrt{3}-2)} = \frac{1}{1-\sqrt{3}+2} = \frac{1}{3-\sqrt{3}} \cdot \frac{3+\sqrt{3}}{3+\sqrt{3}}$$

$$= \frac{3+\sqrt{3}}{9-3}$$

$$= \frac{3+\sqrt{3}}{6} \checkmark$$

$$\#8 \quad m(AP) = \frac{y-0}{x+2}$$

$$m(BP) = \frac{y-2}{x-4}$$

$$\because AP \perp BP:$$

$$\frac{y}{x+2} = \frac{-(x-4)}{y-2} \checkmark$$

$$y(y-2) = -(x-4)(x+2)$$

$$y^2 - 2y = -x^2 + 2x + 8$$

$$x^2 - 2x + y^2 - 2y - 8 = 0 \checkmark$$

$$ii) x^2 - 2x + (1)^2 + y^2 - 2y + (1)^2 = 8 + (1)^2 + (1)^2$$

$$(x-1)^2 + (y-1)^2 = 10 \checkmark$$

$\therefore$ circle

with centre $(1, 1)$
radius $\sqrt{10}$ $\checkmark$

$$b) 3^{2x} - 7 \cdot 3^x - 18 = 0$$

$$\text{let } 3^x = k$$

$$k^2 - 7k - 18 = 0$$

$$(k-9)(k+2) = 0$$

$$k = 9$$

$$3^x = 3^2$$

$$x = 2$$

$$k = -2$$

$$3^x = -2$$

no soln $\checkmark$

$$8c) \quad y = (x-5)^{-2} \quad + \quad y = 4x-3$$

$$y' = -2(x-5)^{-3} \cdot 1 \quad m = 4.$$

$$= \frac{-2}{(x-5)^3}$$

$$\frac{-2}{(x-5)^3} = -\frac{1}{4}$$

$$8 = (x-5)^3$$

$$\sqrt[3]{8} = x-5$$

$$x-5 = 2$$

$$x = 7$$

$$y = 2^2 = \frac{1}{4} \quad \therefore P(7, \frac{1}{4})$$

$$d). \quad \text{let } y = 3ae^x$$

$$u = 3x \quad v = e^x$$

$$u' = 3$$

$$v' = e^x$$

$$\frac{dy}{dx} = 3e^x + 3xe^x$$

$$\int \frac{dy}{dx} = \int 3(e^x + xe^x) dx$$

$$\int_0^2 e^x + xe^x dx = \frac{1}{3} [3xe^x]_0^2$$

$$= [xe^x]_0^2$$

$$= 2e^2 - 0$$

$$= 2e^2$$

9a) $V_{\text{of cylinder}} = \pi r^2 \times h.$

$$= \pi r^2 \times 2x$$

by pythag: $r^2 + x^2 = 100.$

$$r^2 = 100 - x^2.$$

$$\therefore V = 2\pi x(100 - x^2).$$

To max vol: $V' = 0.$

$$V = 200\pi x - 2\pi x^3$$

$$V' = 200\pi - 6\pi x^2 = 0.$$

$$200\pi = 6\pi x^2$$

$$\frac{200\pi}{6\pi} = x^2$$

$$x = \sqrt{\frac{100}{3}}$$

check max: $V'' = -12\pi x < 0.$

$\cap$ max.

b). $F'(x) = e^{3x-2}$

$$f(x) = \frac{e^{3x-2}}{3} + c$$

$$f(1) = \frac{e^{3-2}}{3} + c = \frac{4c}{3}$$

$$c = \frac{4c}{3} - \frac{1}{3}e = e.$$

$$\therefore f(x) = \frac{e^{3x-2}}{3} + e$$

#9c). $\sqrt{x+1} = 5-x$

$$x+1 = 25 - 10x + x^2$$

$$0 = x^2 - 11x + 24$$

$$0 = (x-3)(x-8)$$

$$x=3 \quad \cancel{x=8}$$

$$y=2 \quad B(3,2)$$

or any alternative method.

$$\text{Shaded area} = \int_{-1}^3 (x+1)^{\frac{1}{2}} + \int_3^5 (5-x) dx + \left| \int_5^7 (5-x) dx \right|$$

$$= \frac{2}{3} (x+1)^{\frac{3}{2}} \Big|_{-1}^3 + \left[5x - \frac{x^2}{2} \right]_3^5 + \left| \left[5x - \frac{x^2}{2} \right]_5^7 \right|$$

$$= \frac{2}{3} \sqrt{(x+1)^3} \Big|_{-1}^3 + (25 - 12\frac{1}{2}) - (15 - \frac{9}{2}) + (35 - \frac{49}{2}) - (25 - \frac{25}{2})$$

$$= \frac{2}{3} [8 - 0] + (12\frac{1}{2}) - (\frac{21}{2}) + | \frac{21}{2} - \frac{25}{2} |$$

$$= \frac{16}{3} + 2 + 1 = 2$$

$$= \frac{16}{3} + 4$$

$$= 9\frac{1}{3} \text{ u}^2$$

#10. LHS:

$$(\sec^2 \theta) \sin^2 \theta$$

$$= \frac{1}{\cos^2 \theta} \cdot \sin^2 \theta$$

$$= \tan^2 \theta$$

$$= \text{RHS}$$

$$2 \tan^2 \theta + 5 \tan \theta - 7 = 0$$

$$(2 \tan \theta + 7)(\tan \theta - 1) = 0$$

$$\tan \theta = -\frac{7}{2}$$

$$\theta < 74^\circ 3'$$

$$\tan \theta = 1$$

$$\theta < 45^\circ$$

$$\theta = 105^\circ 57'$$

$$\theta = -74^\circ 3'$$

$$\theta = 45^\circ$$

$$\theta = 225^\circ, -135^\circ$$

10b). $A = \int_{-4}^2 (4-x) - \left(\frac{x^2}{2}\right) dx$ ✓
 $= \left[4x - \frac{x^2}{2} - \frac{x^3}{6} \right]_{-4}^2$ ✓
 $= \left(8 - 2 - \frac{8}{6} \right) - \left(-16 - 8 + \frac{64}{6} \right)$
 $\frac{14}{3} - \frac{-40}{3}$ ✓
 $= 18$ ✓

ii) $P(t, \frac{t^2}{2})$ $2x + y - 4 = 0$.

$$\perp d = \frac{\left| (1)(t) + (1)\left(\frac{t^2}{2}\right) + (-4) \right|}{\sqrt{1^2 + 1^2}}$$

$$\perp d = \frac{\left| t + \frac{t^2}{2} - 4 \right|}{\sqrt{2}}$$

since $-4 \leq t \leq 2$ and d must be positive (or similar) ✓
 $\perp d = \frac{-t - \frac{t^2}{2} + 4}{\sqrt{2}}$ ✓

iii) $A_{\Delta} = \frac{1}{2}(AB) \perp d$.

$$= \frac{\frac{1}{2} 6\sqrt{2} \times (4 - t - \frac{t^2}{2})}{\sqrt{2}}$$

$$= 12 - 3t - \frac{3t^2}{2}$$

$$d(AB) = \sqrt{(-4-2)^2 + (8-2)^2}$$

$$= \sqrt{36 + 36}$$

$$= \sqrt{72}$$

$$= 6\sqrt{2}$$
 ✓

$\therefore$ max area $A' = 0 : -3 - 3t = 0$.

$$-3 = 3t$$

$$-1 = t$$
 ✓

$$\therefore A_{\Delta} = 12 - 3(-1) - \frac{3(-1)^2}{2}$$

$$= 12 + 3 - \frac{3}{2}$$

$$= 13.5$$
 ✓

and 13.5 is $\frac{3}{4}$ of 18. ✓